VSI-Bench Evaluation of Open-Weight VLMs

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1 Setup and protocol

We evaluate three open-weight VLMs on VSI-Bench [Yang et al., 2025], which covers 1316 questions across 288 indoor scenes from ScanNet [Dai et al., 2017] and ScanNet++ [Yeshwanth et al., 2023]. Each model receives 32 frames sampled uniformly from the video scaled to a maximum side length of 448 px and outputs responses using greedy decoding with up to 48 tokens. Prompts are copied from the standard VSI-Bench harness. We report official strict accuracy (ACC) and Mean Relative Accuracy (MRA), and also report a fallback with a lenient parser that extracts numbers written as words or embedded in explanatory text.

Protocol. Evaluated models include Qwen3-VL-8B-Instruct [Bai et al., 2025], Gemma-4-12B-it [Team et al., 2026], and Molmo2-8B [**?**], run on one NVIDIA H100 GPU (PyTorch 2.14.1, transformers 5.22.1 / 4.57.1). Tasks use the full test splits: `route_planning` (Category 1, ACC, n=194), `object_abs_distance` (Category 2, MRA, n=834), and `room_size_estimation` (Category 2, MRA, n=288). We report 95% confidence intervals with resampling at the scene level, which only matters for the `object_abs_distance` task. We also report blind model priors by witholding images and only retrieving model answers based on the prompts.

2 Results

Table 1 summarizes performance across all three tasks. Qwen3-VL leads in every category. Gemma-4 scores near zero under strict evaluation because it responds with sentences rather than direct numbers; the lenient parser extracts these values and shows the resulting scores.

Table 1: Main results (strict score, 95% CI over scenes). Values in parentheses show lenient parsing score when they differ by more than 15 percentage points.

Model	**Route planning** ACC n=194	**Absolute distance** MRA n=834	**Room size** MRA n=288
Qwen3-VL-8B	35.0 [28, 42]	47.9 [45, 51]	57.1 [53, 61]
blind	–	30.6	32.9
Gemma-4-12B	30.4 [24, 37]	0.3 [0, 1] (22.8)	0.0 [0, 0]
blind	–	0.0	0.0
Molmo2-8B	29.4 [23, 36]	0.0 [0, 0] (16.2)	11.5 [8, 14]
blind	–	0.1	32.0

Blind evaluation reveals heavy reliance on priors: Qwen3-VL scores 30.6/32.9 MRA without images, Gemma-4 refuses (0.0), and Molmo2-8B scores higher blind on room size (32.0 vs 11.5) by predicting a constant 20 m^2.

Numeric diagnostics. To evaluate metric estimation, we report the correlation between predictions and ground truth (r), prediction variance relative to ground truth ($\sigma_{\hat{y}}/\sigma_y$), parser failure rate, and median absolute relative error.

Model	Task	r	$\sigma_{\hat{y}}/\sigma_y$	Parse fail.	Med. rel. err
Gemma-4-12B	Abs. dist.	0.2341	0.6624	96.8	54%
Gemma-4-12B	Room size	–	–	100.0	–
Molmo2-8B	Abs. dist.	0.3332	2.9955	99.9	100%
Molmo2-8B	Room size	0.7425	3.2377	38.2	71%
Qwen3-VL-8B	Abs. dist.	0.7143	0.6487	0.0	28%
Qwen3-VL-8B	Room size	0.8136	0.5758	0.0	21%

3 Failure analysis

The cards below show representative failures from all three models: the exact prompt accompanying the 32 frames, the decoded response, and the ground truth. Table 2 categorizes 2789 failures into four mechanisms.

Case 1 — Scale confusion

allenai/Molmo2-8B | Room size | id 533

Prompt. `These are frames of a video. What is the size of this room (in square meters)? If multiple rooms are shown, estimate the size of the combined space. Please answer the question using a single word or phrase.`

Response. `100`

Ground truth. 30.9 m^2

Case 2 — Scale confusion (prior)

Qwen/Qwen3-VL-8B-Instruct | Absolute distance | id 825

Prompt. `These are frames of a video. Measuring from the closest point of each object, what is the distance between the stove and the chair (in meters)? Please answer the question using a single word or phrase.`

Response. `1.7`

Ground truth. 3.6 m

Case 3 — Route planning

Qwen/Qwen3-VL-8B-Instruct | Route planning | id 4961

Prompt. `These are frames of a video. You are a robot beginning at the sink and facing the heater. You want to navigate to the doorframe. You will perform the following actions (Note: for each [please fill in], choose either 'turn back,' 'turn left,' or 'turn right.'): 1. [please fill in] 2. Go forward until the doorframe. You have reached the final destination. Options: A. Turn Back B. Turn Left C. Turn Right Answer with the option's letter from the given choices directly.`

Response. `B`

Ground truth. A. Turn Back

Case 4 — Refusal / abstention

google/gemma-4-12B-it | Room size | id 530

Prompt. `These are frames of a video. What is the size of this room (in square meters)? If multiple rooms are shown, estimate the size of the combined space. Please answer the question using a single word or phrase.`

Response. `I cannot determine the size of the room from the provided images.`

Ground truth. 26.4 m^2

Case 5 — Answer format

google/gemma-4-12B-it | Absolute distance | id 682

Prompt. `These are frames of a video. Measuring from the closest point of each object, what is the distance between the tv and the sofa (in meters)? Please answer the question using a single word or phrase.`

Response. `The distance between the tv and the sofa is 2.15 meters.`

Ground truth. 2.1 m (0.00 strict / 1.00 lenient)

Table 2: Failure taxonomy. Near misses within 30% relative error are excluded, since MRA awards partial credit there.

Class	n	Mechanism
Scale confusion	1824	large numeric error; predictions drift towards the training priors
Egocentric direction	378	incorrect multiple choice; models favor options B and C over A and D
Refusal / abstention	355	model outputs refusal text instead of estimating values
Answer format	232	correct value produced, but the benchmark parser fails to parse it

4 Discussion

Qwen3-VL-8B demonstrably uses video imagery: In the blind evaluation, it guesses numbers from its training distribution (31% MRA on both tasks), and adding imagery raises MRA by 17 and 24 points. This suggests it may have been pretrained on VSI-Bench or similar data.

Gemma-4-12B avoids guessing when blind, which is desirable, but also abstains on all room-size tasks even with video and ignores instructions to output only numbers. A lenient parser is thus needed to extract its answers ($0.3 \rightarrow 22.8$).

Molmo2-8B's predictions correlate with room size ($r = 0.74$) but $3\times$ more variance than the ground truth. With blind evaluation, it emits a constant 20 m^2 (32.0 MRA). Like Gemma-4, it also struggles with instruction following and outputs many unparsable responses.

All three models perform poorly on route planning (29.4–35.0% ACC), partly because they are biased towards B and C regardless of camera motion. Overall, the models are poor at predicting absolute metric dimensions, and require grounding with 3D-supervision in the style of VLM-3R [Fan et al., 2026] to become more accurate.

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